Midterm 2 Review

ECON 441: Introduction to Mathematical Economics

Div Bhagia

What's on the exam

- Wednesday, October 21, in class. 60 minutes, 20 points.
- Module III: Calculus (Lectures 8–13)
- Chiang & Wainwright: 6.2–6.4, 6.7, 7.1–7.4, 8.1, 8.2, 8.4, 8.5, 10.5, 14.1–14.3
- The help sheet provided with the exam is on the course website
- Closed book: no phones or computers

The Derivative

Average rate of change, then let the change shrink:

$$\frac{dy}{dx} = f'(x) = \lim_{\Delta x \to 0} \frac{f\,(x + \Delta x) - f\,(x)}{\Delta x}$$

Use the limit definition to find $f'(x)$ for $y = f(x) = x^2 + 3x$.

Limits and Continuity

$\lim_{x \to a} f(x) = L$ exists only if the left-side and right-side limits agree.

f is **continuous** at a if the limit exists and $\lim_{x \to a} f(x) = f(a)$.

$$f(x) = \begin{cases} x + 1 & \text{if } x < 1 \\ 3 & \text{if } x = 1 \\ x + 1 & \text{if } x > 1 \end{cases}$$

Does $\lim_{x \to 1} f(x)$ exist? Is f continuous at $x = 1$?

Continuity and Differentiability

Continuity is **necessary** for differentiability, but not **sufficient**.

$$g(x) = \begin{cases} 2x & \text{if } x < 1 \\ x + 1 & \text{if } x \geq 1 \end{cases}$$

Plot g. Is it continuous at $x = 1$? Is it differentiable there?

The Rules

Constant, power, sum-difference; then

$$\frac{d}{dx}[fg] = fg' + f'g \qquad \frac{d}{dx}\frac{f}{g} = \frac{f'g - fg'}{g^2} \qquad \frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}$$

Find dy/dx:

- $y = (x^2 + 1)(3x - 2)$
- $y = \dfrac{x}{x + 2}$
- $y = (2x + 1)^3$

Exponential and Log Functions

$$\frac{d}{dx}e^{f(x)} = f'(x)e^{f(x)} \qquad \frac{d}{dx}\ln f(x) = \frac{f'(x)}{f(x)}$$

$$\ln(uv) = \ln u + \ln v \qquad \ln(u/v) = \ln u - \ln v \qquad \ln u^a = a \ln u$$

Find dy/dx:

- $y = 3e^{2x}$
- $y = x^2 \ln x$
- $y = \ln(x^2 + 4)$

Elasticity in Logs

$$\varepsilon = \frac{dy}{dx} \cdot \frac{x}{y} = \frac{d \ln y}{d \ln x}$$

Demand satisfies $\ln Q = 2 - 0.5 \ln p$. What is the price elasticity of demand? Is demand elastic or inelastic?

Partial Derivatives

Differentiate with respect to one variable, holding the others constant.

$$f(x, y) = x^2 y + y^3$$

Find f_x, f_y, and the gradient ∇f. Is f_x increasing in y?

Production Function

$$Q = K^{0.3} L^{0.7}$$

Find the marginal product of capital Q_K and of labor Q_L.

Is the marginal product of labor decreasing in L?

Elasticity

$$\varepsilon = \frac{\text{Percentage change in } y}{\text{Percentage change in } x} = \frac{dy}{dx} \cdot \frac{x}{y}$$

Demand is $Q = 50 - 2p$. Find the elasticity at $p = 10$. Elastic or inelastic?

Total Differential and Total Derivative

For $y = f(x_1, x_2)$:

$$dy = f_1\, dx_1 + f_2\, dx_2 \qquad \frac{dy}{dx_1} = f_1 + f_2 \frac{dx_2}{dx_1}$$

$z = x^2 + xy$, where $x = t$ and $y = t^2$. Find dz and dz/dt.

Total Derivative with Time

Output $Y(t) = K(t) + N(t)$, where $K(t) = 3t$ and $N(t) = e^{0.5t}$.

Find the rate of change of output over time, dY/dt.

Implicit-Function Rule

For $F(x, y) = 0$, totally differentiate: $F_x\, dx + F_y\, dy = 0$, so

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

$F(x, y) = x^2 + y^2 - 25 = 0$. Find dy/dx, and its value at $(3, 4)$.

Market Equilibrium

Demand $Q^d = 60 - 3P + 0.1Y$, supply $Q^s = 10 + 2P$.

Write the equilibrium condition as $F(P, Y) = 0$ and find $\partial P/\partial Y$ by the implicit-function rule. Check by solving for P.

Rules of Integration

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + c \ (n \neq -1) \qquad \int e^x \, dx = e^x + c \qquad \int \frac{1}{x} \, dx = \ln x + c$$

Find:

- $\int (3x^2 + 2x) \, dx$
- $\int 4e^{-2x} \, dx$

Definite Integrals

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Evaluate $\int_1^2 \left(x + \frac{1}{x}\right) dx$

Marginal to Total

A firm's marginal cost is $MC(Q) = 4Q + 1$ and its fixed cost is 20.

Find the total cost function $C(Q)$. What is the cost of raising output from $Q = 2$ to $Q = 4$?

For your review

- Lecture slides and worksheets
- Practice problems 8–13 with solutions, also collected as one PDF for the module
- Module notes for Calculus and for Log and Exponential Functions
- Sample exam and help sheet, on the Exams page
- Additional problems in the textbook

Good luck!